QUINCY RYAN

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Davis, CA (contact for full address)

EDUCATION

University of California, Davis *Fall 2023–May 2026*
B.S. in Cognitive Science with a Computational emphasis. Minors in Neuroscience and Music
Dean's List Fall 2023
Citation for Outstanding Performance in Cognitive Science 2025
Phi Beta Kappa Honor Society (Joined Spring 2026)
GPA: 3.827/4.0 (UC Davis)

Pasadena City College *2019–2023*
Honors for Superior Achievement in Philosophy (2021)
Dean's Honors (Spring 2020, Spring 2023)

California Institute of the Arts *2017–2019*
Music Performance and Composition

RESEARCH & TEACHING EXPERIENCE

Cognitive Control Lab: Dr. Xiaomo Chen October 2025–Present
Research Assistant *UCD: Center for Neuroscience*

- Research mentors: Orhan Soyuhos & Brinda Sevak.
- Programmed and deployed video-based eyetracking and neuro-physiology paradigm for non-human primates.
- Analyzing eye-tracking data in meaning-map tasks using linear mixed-effects models (Bambi/PyMC) to identify behavioral and neural-mechanistic correlates.

Cognition in Context Lab. PI: Dr. Yuko Munakata. January 2025–May 2026
Research Assistant *UCD: Center for Mind and Brain*

- Research mentor: Dr. Jesse Niebaum.
- Designed and implemented complex cognitive and behavioral experiments using PsychoPy and the EYNA eye-tracking system.
- Analyzed behavioral and eye-tracking data in R and Python.
- Ran cognitive and behavioral experiments with adults and children.
- Discussed relevant literature and experimental design.
- Acquired CITI Social & Behavioral Research Basic course certification.
- Assisted with participant recruitment and scheduling.

Computational Cognitive Neuroscience Textbook: Dr. Randall O'Reilly, Dr. Munakata Jan 2025–Sept 2025
Programmer *CogentCore/CompCogNeuro/UCD*

- Improving user experience of the CompCogNeuro's lab simulations.
- Programming in Go using the Cogent Core Framework
- CompCogNeuro Repository: github.com/CompCogNeuro/sims
- Lab Simulations: compcogneuro.org/simulations

Neuro-Tech Davis: Attention detection BCI November 2024–March 2025
Project Manager *UCD*

- Led and taught a team of 6 undergraduate students in research, scientific writing, experimental design, and programming.
- Managed the development of a BCI intended to detect and classify attention/focus (EEG, ML) and refocus the user through adaptive stimuli.

Janata Lab: Dr. Petr Janata August 2024–April 2025
Research Assistant *UCD: Center for Mind and Brain*

- Sole software developer for "Brainmuse" project

- Integrated, refactored, and developed multiple models of music cognition
- Used C++, Bash, Python, the JACK audio framework, and other development tools

Academic Volunteer work Spring 2026, Winter–Spring 2024
Student Assistant *UCD*

- Dr. Xiaomo Chen and Dr. Gregg Recanzone's NPB 110 Neurobiology: Assisting students with course work related to foundations of molecular, cellular, systems, and cognitive neuroscience.
- Professor Butner's ECS 032a and 036a: Assisting students with coursework in C and Python. Supporting programming lab sessions and helping students debug code

INDUSTRY EXPERIENCE

CD International 2026–Present
Software Developer

- Built Mogent, a Go CLI that composes agent instructions and skills from reusable, versioned libraries, with drift detection across generated outputs, disk, and upstream sources.
- Designed advanced LLM agent tooling: multi-agent orchestration, shared skill libraries, and handoff/continuity workflows for long-running agent work.
- DevOps and Nix administration: flakes, pinned toolchains, reproducible dev shells, CI, and declarative NixOS / nix-darwin configuration across systems and repos.

TECHNICAL SKILLS

Scientific Computing: Python (ML, PsychoPy, Bambi/PyMC LMMs), R, MATLAB
Systems: Rust, Go, C, C++
Development Tools: Git/GitHub, Bash scripting, Unix CLI tools, Nix flakes, direnv, Python dependency management via Poetry and uv
Typesetting & Document Preparation: Markdown, LaTeX
Operating Systems: Unix, Linux (Ubuntu, Fedora, NixOS), macOS (Homebrew, Nix Darwin), Windows (WSL), VMs
Hardware: Soldering, computer assembly and repair, audio/video processing and I/O, carpentry and fabrication

RESEARCH INTERESTS

- Visual attention and meaning guidance during complex scene viewing; eye-tracking methodologies.
- Computational and neural mechanisms of scene attention and comprehension.
- Social cognition, theory of mind, and individual differences in social attention.
- Assistive and translational applications of attention research.

RELEVANT COURSEWORK

Neuroscience & Cognitive Science:
Computational Cognitive Neuroscience, Cognitive Neuroscience, Systems Neuroscience, Neural Mechanisms of Behavior, Theoretical Neuroscience Seminar in Neural Networks (graduate level), Evolution of Mind, Neurobiology of Mental Illnesses, Brain Disorders

Computer Science & Mathematics:
Machine Learning, Discrete Math for Computer Science, Object-Oriented Programming, Data Structures, Calculus, Linear Algebra, Differential Equations, Computational Linguistics, Theory of Computation

Psychology & Research Methods:
Research Methods in Psychology, Cognitive Psychology, Introduction to Biopsychology, Social Cognition, Introduction to Linguistics

PRESENTATIONS

Recurrent Neural Networks: Learning Rule Derivations *Fall 2025*
NPB 287A: Theoretical Neuroscience Seminar (graduate level)
Presented mathematical derivations of recurrent network learning rules